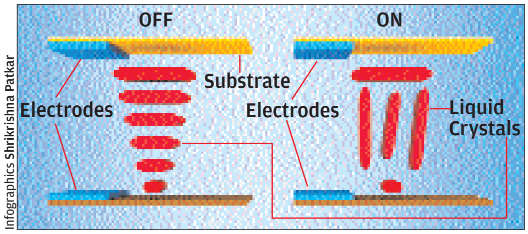


Figure 1: Orientation of liquid crystals in a TN panel. Note the unorthodox alignment in the On position

Due to the significantly higher costs involved, S-IPS technology is seen in larger LCD panels, 20-inch and above

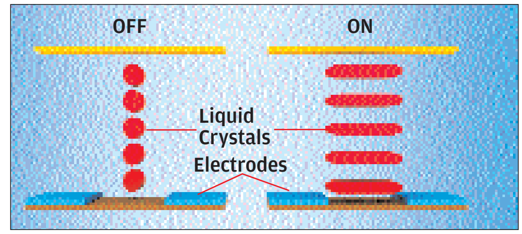


Figure 2: A view of the liquid crystals between the glass substrates in an IPS panel. Note the way the crystals stretch when electric current is applied—in-plane

IPS

The second-type of LCDs we're looking at are the much costlier IPS (In-Plane Switching) panels. Developed by Hitachi in 1996, IPS panels were intended to remedy all the problems associated with TN panels mentioned above. Incidentally, IPS panels are natively 8-bit, which means that they can display the entire range of 16.7 million colours without the need for any dithering whatsoever. This makes them a good choice for professionals working with image and graphics applications. A newer version of IPS is S-IPS (Super IPS), which addresses the inherently poor response times that plagued first-generation IPS panels.

How IPS Works

As the name suggests, the crystals in the panel do not change orientation during off and on operations, remaining parallel to the panel's plane. Notice the elongation of the liquid crystals as they switch to their active state. Also notice the position of the electrodes—they're on the same wafer. This design is more space-consuming as well, which leads to one of the major shortcomings of IPS panels—a poor contrast ratio which causes relatively lower brightness levels. The benefits of IPS technology are the generously wide viewing angles and the brilliant colour reproduction.

Due to their original design, IPS panels suffered from very slow pixel response times, which made them unsuitable for gaming and multimedia applications because of all the *ghosting*.

A shot in the arm for IPS came in 1998 when Hitachi Corporation developed S-IPS or Super IPS. S-IPS significantly lowered pixel response times from 50 ms (for IPS) to about 25 ms. Later, in a joint venture, LG and Philips worked out several chinks in the S-IPS armour, further reducing response times to about 16 ms. None of this affected the brilliant image quality, wide viewing angle and colour reproduction, thus making S-IPS panels the only choice for the discerning.

Due to the significantly higher costs involved in their manufacture, S-IPS technology is reserved for larger panels, above 20-inches.

MVA

MVA (Multi-Domain Vertical Alignment) panels came into existence sometime in 1998, thanks to Fujitsu, as a compromise between TN and IPS technologies. TN panels offered superb response

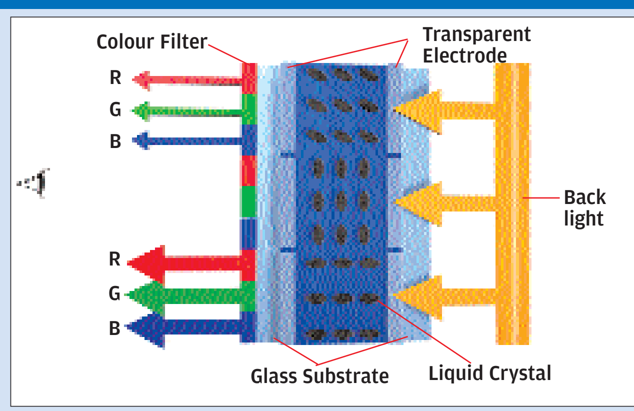
How An LCD Works

Liquid crystal display panels are named after the semi-fluid crystals present in them. Liquid crystals have a couple of peculiar properties that made flat panels possible.

Firstly, they change shape when electric current is applied to them. Different types of panels (as mentioned above) will have crystals taking on differing shapes when an electrical current is applied, but one thing stays common—a change occurs. Let's not get into the types of liquid crystals around.

Secondly, liquid crystals can transmit polarised light. An LCD basically works on the principle of alternatively blocking and allowing light to pass through. This blocking and allowing is done by the liquid crystals themselves as they change structure when an electric current is passed through them.

As you can see, the liquid crystals are sandwiched between two layers of polarised glass. The inner sides of the glass panels, the sides that aren't polarised, have microscopic grooves hewn into them. These grooves are made by a special sort of polymer which is conductive, and each groove corresponds to where a pixel's alignment would be, and the grooves are perpendicular to each other. Liquid crystal material is placed between these two layers of glass substrate, and a polarising film is put on the outer side of each glass panel. The individual rows and columns inside the substrate created by the polymer are hooked up to integrated circuits. This is how



current is supplied to a particular row or column to activate a particular pixel.

Now, suppose a current is passed through this assembly to a particular pixel; the voltage is received by the liquid crystals causing them to change shape. Now this change of shape differs according to the type of panel being used, but basically, it alters the intensity of light passing through the liquid crystals onto the pixel.

Note that the liquid crystals merely allow or disallow light being created by the polarised panels to pass through them. They do not create any light of their own.